

CoRoT-2b

R-1274

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_190623 · created 2026-07-31T14:14:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458657.7803 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.064
Transit depth (Rp/Rs)^2	4.46 +/- 2.69 [%]
Orbital Inclination (inc)	83.89 +/- 2.06
Airmass coefficient 1 (a1)	1.18 +/- 0.56
Airmass coefficient 2 (a2)	-0.102 +/- 0.091
Scatter in the residuals of the lightcurve fit is	47.64 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.077 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.064 vs 0.1667 (deviation 0.69σ)
Duration fit vs published	0.077 d vs 0.095 d (deviation 0.9σ)
Mid-time O–C	4.9 min
Depth SNR	3.3
Residual scatter	47.64 %

Light curve

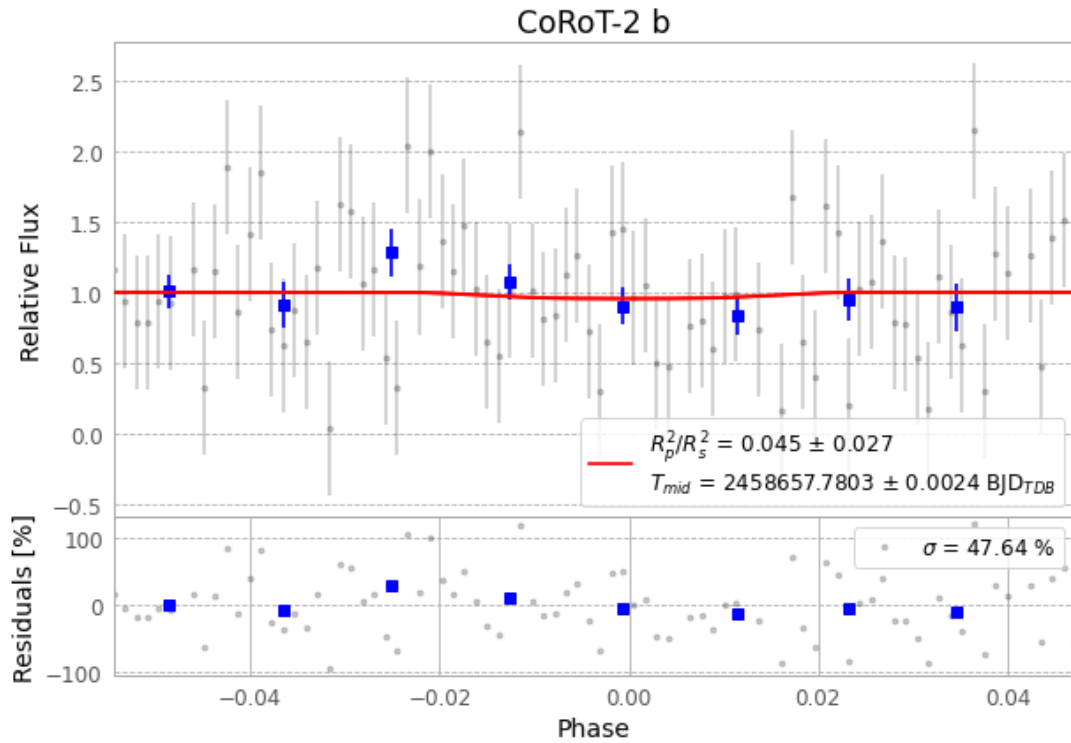


Figure 1 — FinalLightCurve_CoRoT-2 b_2019-06-22.png (SHA-256 f1ac1a49a8f76541...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [85, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 712b1058cf689c16...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), CoRoT-2190623041812.FITS → CoRoT-2190623083312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-190623.log (b66d479ea176...), FinalLightCurve_CoRoT-2 b_2019-06-22.png (f1ac1a49a8f7...), FinalLightCurve_CoRoT-2 b_2019-06-22.csv (3989565f9e53...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:14Z.